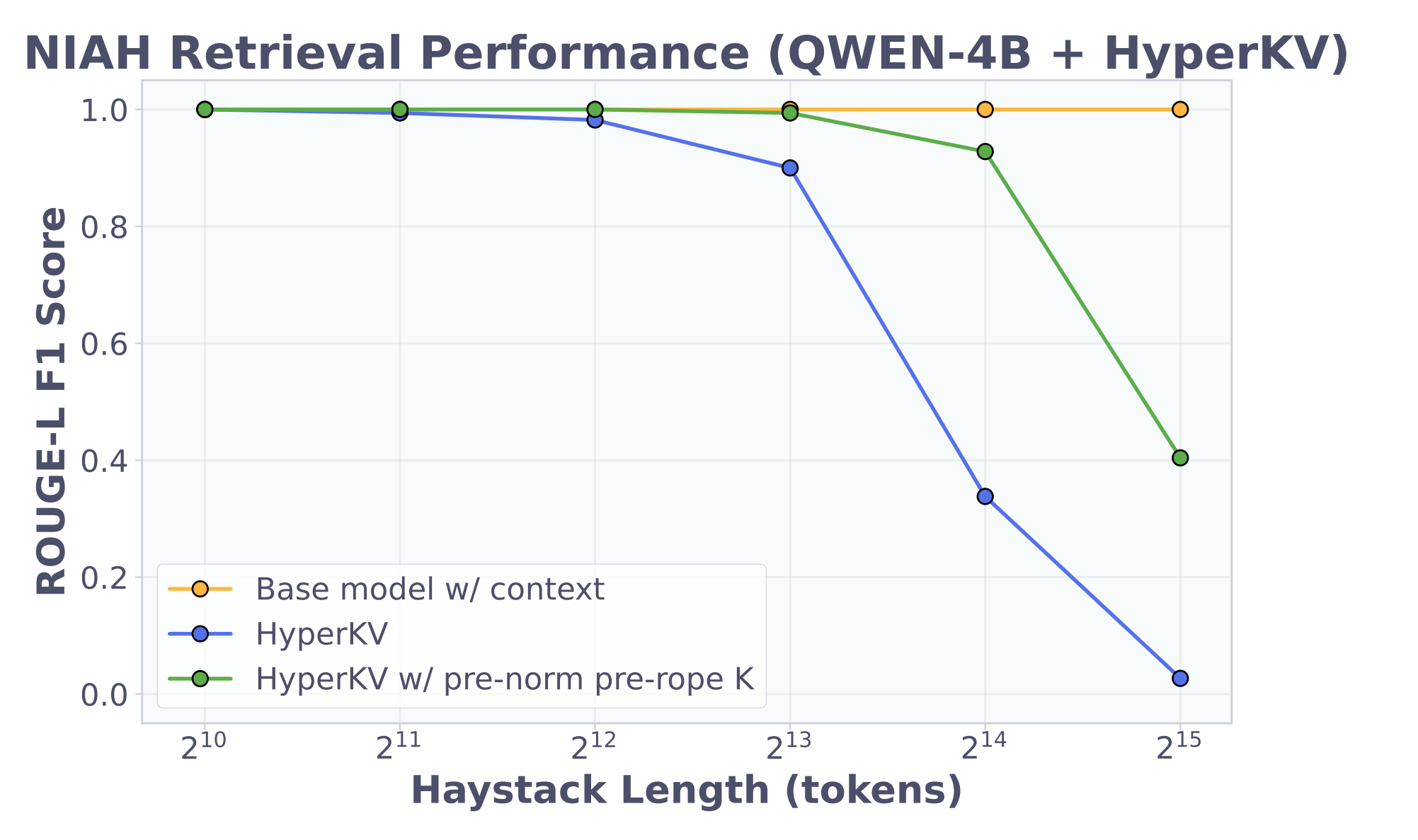

NIAH Retrieval Performance (QWEN-4B + HyperKV)
ROUGE-L F1 Score
1.0
0.8
0.6
0.4
0.2
0.0
Base model w/ context
HyperKV
HyperKV w/ pre-norm pre-rope K
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}
Haystack Length (tokens)